

The Greek philosopher Aristotle produced the first classification of animals and separated vertebrates from invertebrates.

In 2001, scientists published the results of the Human Genome Project: a catalog of all human genes.



2015

Artificial parts

False limbs had been used since antiquity, but the 20th century brings more sophisticated artificial body parts. The first bionic eye is implanted in 2015.



2010s

Fossil evidence

More discoveries of ancient creatures, often preserved in amber, lead scientists to new conclusions, such as the realization that many dinosaurs had feathers.



2017

Changing genes

In the late 20th century scientists become able to edit the genes of living things. In 2017, some mosquitoes are genetically altered to try and stop the spread of the disease malaria.

1970-PRESENT

1930s



The rise of ecology

The study of ecology (how organisms interact with their surroundings) emerges in the 1930s, as British botanist Arthur Tansley introduces the idea of ecosystems.

1928

Antibiotics discovered

British biologist Alexander Fleming discovers that a substance—penicillin, the first known antibiotic—stops the growth of microbes. Antibiotics are now used to treat many bacterial infections.



1900s



Chromosomes and genes

American scientist Thomas Hunt Morgan carries out experiments on fruit flies, which prove that the units of inheritance are carried as genes on chromosomes.

1859

Evolution

British biologist Charles Darwin publishes a book—*On the Origin of Species*—explaining how life on Earth has evolved by natural selection.

CHARLES DARWIN



1860s

Microbes

An experiment by the French biologist Louis Pasteur proves microbes are sources of infection. It also disproves a popular theory which had argued that living organisms could be spontaneously generated from nonliving matter.



MICROBE

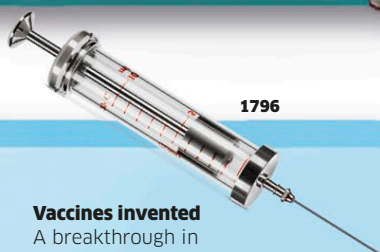
Early 20th century

Better microscopes and advances in studying the chemical makeup of cells helped to show how all life carries a set of building instructions—in the form of chromosomes and DNA—while ecology and behavior became new topics of focus.

20TH CENTURY

1800-1900

1796

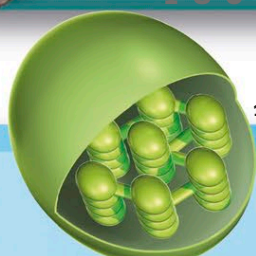


Vaccines invented

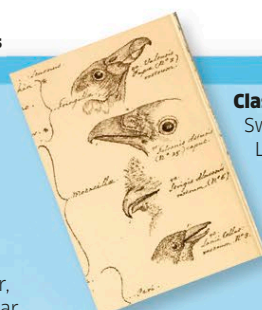
A breakthrough in medicine, the first vaccine is used by British doctor Edward Jenner to protect against a deadly disease—smallpox.

Photosynthesis discovered

In the 1770s, the experiments of a Dutch biologist, Jan Ingenhousz, show that plants need light, water, and carbon dioxide to make sugar.



1770s



1735

Classifying life

Swedish botanist Carl Linnaeus devises a way of classifying and naming plants and animals that is still used today.

VAN LEEUWENHOEK'S MICROSCOPE

Microscopic life

British scientist Robert Hooke views cells down a microscope and inspires a Dutchman, Antony van Leeuwenhoek, to invent his own unique version of a microscope.

1665



1600-1800

Cataloging life

The ancient Greeks are the first to try classifying life, but it is not until the 16th century that species are first cataloged in large volumes.



17TH CENTURY

17th-18th centuries

New scientific experiments added to the wealth of knowledge laid down by the first philosophers. This research helped to answer important questions about life's vital processes, such as blood circulation in animals and photosynthesis in plants.

1628

Blood circulation

A British doctor called William Harvey combines observation with experiment to show how the heart pumps blood around the body.

